COMPUTATIONAL INTELLIGENCE | MRCET

(CSE – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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IV Year B. Tech-II Semester Major Project Summary Sheet

Project Title:	HireLens AI – Explainable AI Resume Screening System				
Project Code:	R22A6695	**Batch Size:**	03	**Batch:**	2022-26
Domain/Area:	Natural Language Processing (NLP) and Explainable AI (XAI)	**SDG Mapping:**	Decent Work and Economic Growth		

Abstract:

HireLens AI is an Explainable Artificial Intelligence–based resume screening system designed to make recruitment faster, transparent, and more data-driven. Most existing Applicant Tracking Systems (ATS) work as black boxes, providing hiring decisions without any reasoning, which creates confusion for both candidates and recruiters. HireLens AI solves this by using Natural Language Processing (NLP) and machine learning to extract key details like skills, education, work experience, projects, and certifications from resumes, then comparing them against job descriptions to generate a compatibility score along with clear, interpretable explanations of matched skills, missing qualifications, and relevant experience. It also includes a side-by-side candidate comparison feature to help recruiters make better-informed decisions. The expected outcome is an AI-powered recruitment tool that brings efficiency and fairness to the hiring process, helping recruiters choose the right candidates while giving applicants clear insight into their evaluation.

Technical (S/w & H/w) Specifications

Software Specifications

- Frontend: React 19, Vite 8.0, TailwindCSS
- Python 3.10+, FastAPI, SQLAlchemy
- NLP: spaCy, sentence-transformers, SHAP
- DB: SQLite via SQLAlchemy ORM

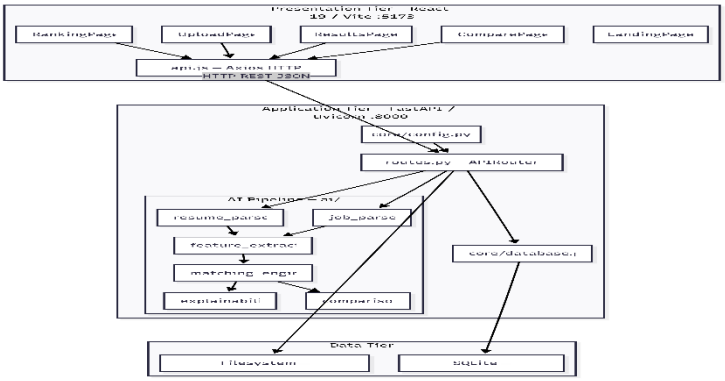
Hardware Specifications

- Processor: Intel i5 (8th gen) or above
- Memory: 8GB or above
- Hard Disk: 5GB free space (SSD preferred)
- Internet Connectivity

Module(s) Specifications

Module 1: Resume Upload & Parsing
Module 2: Job Description Submission & Parsing
Module 3: AI Feature Extraction & Scoring Engine
Module 4: Explainability & Report Generation
Module 5: Batch Processing & Candidate Ranking
Module 6: Candidate Comparison & UI/UX Dashboard

Architecture Diagram



Methodology

- Analysis: Identified the lack of transparency in ATS hiring decisions and the need for explainable AI in recruitment.
- Design: Planned a six-stage XAI pipeline with weighted scoring, NLP-based parsing, and a React frontend.
- Development: Built resume parser, job parser, matching engine, explainability module, and React UI with FastAPI backend.
- Testing: Verified match score accuracy, explanation quality, and UI responsiveness across different resume formats.
- Deployment: Deployed locally using Uvicorn and Vite dev server; tested end-to-end with sample resumes and job descriptions.

Existing System

- Commercial ATS platforms (Workday, Greenhouse) operate as black boxes with no explanation for hiring decisions.
- Keyword-based filters cause false rejections for synonymous skill terms.
- No multi-dimensional scoring or candidate comparison in existing tools.

Proposed System

- Explainable AI pipeline with transparent, human-readable justification for every screening decision.
- Weighted multi-dimensional scoring across skills, experience, education, and projects.
- Batch processing, ranked leaderboard, and side-by-side candidate comparison with hiring recommendation.

Guide Details

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